

Distinct Distances in a Cubic Lattice

Computation of A275672 and Explicit Asymptotic Bounds

Computational and analytic research report

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Abstract

Let a_n be the maximum size of a subset of $\{0, \dots, n-1\}^3$ whose unordered pairwise Euclidean distances are all different. We develop an exact search based on squared-distance colors, symmetry classes of a unique longest edge, dynamic compatibility graphs, and arithmetic capacity constraints. The computational results include the new exact terms $a_7 = 10$ and $a_8 = 12$, together with constructions of 13 and 14 points in cubes of side lengths 9 and 10 and a certificate catalogue through side length 30. Explicit point sets, complete case records where available, and programs for independent verification accompany this report.

For the asymptotic problem we give a positive-definite Gaussian-kernel method. A fully analytic three-atom certificate proves $\limsup a_n/n \leq 1.8712786400\dots$. A four-atom certificate, checked with integer arithmetic and a rigorous interpolation bound, improves this to

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \sqrt{\frac{1}{6\kappa}} = 1.848895345708975\dots < 1.849, \quad \kappa = \frac{48755553510102913673137}{10^{24}}.$$

We also derive exact finite upper bounds from geometric series, and give an elementary alteration proof explaining the exponent $2/3$ in the known lower-bound theory. The classical logarithmically stronger lower bound is $\Omega(n^{2/3}(\log n)^{1/3})$. Neither the true growth exponent nor strict monotonicity is settled.

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1 The problem and the results

1.1 Normalization and the meaning of an exact term

Write

$$G_n = \{0, \dots, n-1\}^3, \quad a_n = \max\{|A| : A \subseteq G_n, \|x-y\|^2 \text{ are distinct over } \{x, y\} \in \binom{A}{2}\}.$$

Translation identifies this convention with the question's $\{1, \dots, n\}^3$. We set $a_0 = 0$. Squared distances are positive integers, so all finite computations can avoid square roots and floating-point equality tests.

The condition concerns *all unordered pairs*. In particular, equal lengths sharing a vertex are forbidden, as are equal lengths on disjoint edges. Checking only distinct displacement vectors, or only the absence of isosceles triangles, would solve a different problem.

The Math.StackExchange question dates from 2016, and the corresponding OEIS entry still lists

$$(a_0, \dots, a_6) = (0, 1, 3, 4, 6, 7, 9)$$

at the time of retrieval [2, 1]. The entry gives intervals

$$10 \leq a_7 \leq 12, \quad 11 \leq a_8 \leq 13, \quad 12 \leq a_9 \leq 15, \quad 13 \leq a_{10} \leq 17.$$

These are the baseline against which the finite results below are compared.

A point list proves a lower bound once its distances have been checked. A failed heuristic run proves no upper bound. An exact new term requires both a valid point list and a complete impossibility argument for the next size. This distinction is maintained throughout the tables, the data files, and the search output.

1.2 Finite results

The exhaustive computations and integer-coordinate witnesses give the following improvements over the retrieved OEIS entry. A catalogue of integer-coordinate witnesses is supplied for every side

Table 1: Advances beyond the retrieved OEIS baseline.

n	Published interval	Conclusion here	Evidence
7	$10 \leq a_7 \leq 12$	$a_7 = 10$	Witness and exhaustive search
8	$11 \leq a_8 \leq 13$	$a_8 = 12$	Witness and exhaustive search
9	$12 \leq a_9 \leq 15$	$13 \leq a_9 \leq 15$	New witness
10	$13 \leq a_{10} \leq 17$	$14 \leq a_{10} \leq 17$	New witness

length from 7 through 30. The larger examples are lower bounds unless the table explicitly labels an exact value.

1.3 Asymptotic results

The established lower bound of Lefmann and Thiele is

$$a_n \geq c n^{2/3} (\log n)^{1/3} \quad \text{for all sufficiently large } n, \tag{1}$$

where $c > 0$ is an absolute constant [3, 4]. Their variable is the *side length*; the grid has n^3 points. This convention matters when comparing exponents in the literature.

The main upper-bound conclusions of this report are

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq C_6 := \left[\frac{5(1 - 6^{-12})}{12 \log 6} \left(\frac{121}{49} \right)^3 \right]^{1/2} = 1.8712786400 \dots, \quad (2)$$

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq C_{\text{cert}} := \sqrt{\frac{1}{6\kappa}} = 1.8488953457 \dots < \frac{1849}{1000}. \quad (3)$$

The first bound has a short analytic proof. The second uses a small, fully specified integer-arithmetic certificate. In addition to its asymptotic implication, the same certificate supplies computable upper bounds for each finite n ; see section 10.

These bounds improve the elementary coefficient $\sqrt{5}$. They do not improve the exponent 1 in the upper bound, and they do not imply $a_n = o(n)$. The Gaussian certificates are proved here; this report does not claim that a complete publication-priority search has established their novelty. Larger-grid witnesses are reported as certified constructions, without claiming global record status.

1.4 A recent result that cannot be transferred to dimension three

The 2026 preprint of Croot, Mao, Pohoata, Sheffer, and Yip proves the planar upper bound

$$f_2(n) \ll n \exp\left(-c \frac{\log n}{\log \log n}\right)$$

and treats positive-definite *binary* quadratic forms [5]. It supplies valuable motivation for arithmetic methods, but it does not prove the analogous result for the ternary form $x^2 + y^2 + z^2$. A projection of a distance-distinct cubic configuration need not remain distance-distinct. Thus neither that theorem nor projection justifies a sublinear upper bound for a_n .

2 The exact set of available squared distances

Define the squared-distance palette

$$\mathcal{D}_n = \{u^2 + v^2 + w^2 : 0 \leq u, v, w \leq n-1\} \setminus \{0\}, \quad d_n = |\mathcal{D}_n|.$$

Every member of $\mathcal{D}_n$ occurs between two points of the entire cube: take 0 and (u, v, w) . This does *not* mean that all members can occur simultaneously in a small distance-distinct subset.

Proposition 2.1 (Palette bound). *For $n \geq 1$,*

$$\binom{a_n}{2} \leq d_n, \quad a_n \leq U_n^{\text{pal}} := \left\lfloor \frac{1 + \sqrt{1 + 8d_n}}{2} \right\rfloor.$$

Proof. The map from unordered pairs of the selected set to their squared distances is injective into $\mathcal{D}_n$. □

Direct generation needs only the triples $0 \leq u \leq v \leq w < n$. It takes $O(n^3)$ elementary iterations and a bit array of $O(n^2)$ bits. The delivered implementation instead uses Boolean polynomial products encoded as arbitrary-precision integers. With

$$B = \bigvee_{j=0}^{n-1} 2^{j^2}, \quad B_2 = \bigvee_{j=0}^{n-1} (B \ll j^2), \quad B_3 = \bigvee_{j=0}^{n-1} (B_2 \ll j^2),$$

the set bits of B_3 , other than bit zero, are exactly $\mathcal{D}_n$. Here $\vee$ means bitwise OR. No arithmetic convolution carries occur. There are $2n$ big-integer shift/OR operations on $O(n^2)$ bits; on a word model their cost is $O(n^3/w)$ word operations, where w is the word length. This makes palette computations to side length 1000 straightforward in the supplied script.

Remark 2.2. The coefficient support of $(1 + z + z^4 + \cdots + z^{(n-1)^2})^3$ is another exact description. Ordinary polynomial convolution can be accelerated with exact transform methods, but coefficients must be reconstructed without modular cancellation. A numerical FFT followed by unqualified rounding would not, by itself, be an exact certificate.

3 Arithmetic capacity constraints

3.1 Checkerboard parity and a short improvement at side length seven

For a lattice point x , put $\epsilon(x) = x_1 + x_2 + x_3 \pmod{2}$. Since $z^2 \equiv z \pmod{2}$,

$$\|x - y\|^2 \equiv \epsilon(x) + \epsilon(y) \pmod{2}.$$

Let o_n and e_n be the numbers of odd and even members of $\mathcal{D}_n$.

Proposition 3.1 (Parity cut bound). *If a distance-distinct set has size m , some integer s , $0 \leq s \leq m$, satisfies*

$$s(m - s) \leq o_n, \quad \binom{s}{2} + \binom{m - s}{2} \leq e_n. \quad (4)$$

Proof. Partition the selected points by ϵ . Precisely the pairs crossing the partition have odd squared distance. All their squared distances must be different, and likewise for the pairs within the parts. $\square$

For $n = 7$, the exact palette contains 66 elements, 30 odd and 36 even. If $m = 12$, all 66 colors would have to be used, forcing

$$s(12 - s) = 30, \quad \text{equivalently} \quad (s - 6)^2 = 6.$$

This is impossible for integral s , and proves $a_7 \leq 11$ before any extensive search. It is a useful illustration of how a modest arithmetic relaxation can beat the unstructured color count.

3.2 Coordinate parity modulo four

Let z_α count selected points with coordinate parity $\alpha \in \{0, 1\}^3$. If $h(\alpha, \beta)$ is Hamming distance, then

$$\|x - y\|^2 \equiv h(\alpha, \beta) \pmod{4}.$$

Indeed, an even coordinate difference contributes 0 modulo 4 and an odd coordinate difference contributes 1.

For any allowed set of colors $D \subseteq \mathcal{D}_n$, put $b_r(D) = |\{d \in D : d \equiv r \pmod{4}\}|$. Necessary constraints are

$$\sum_{\alpha} z_{\alpha} = m, \quad \sum_{\alpha} \binom{z_{\alpha}}{2} \leq b_0(D), \quad (5)$$

$$\sum_{\substack{\alpha < \beta \\ h(\alpha, \beta) = r}} z_{\alpha} z_{\beta} \leq b_r(D), \quad r = 1, 2, 3. \quad (6)$$

The ordering $\alpha < \beta$ merely avoids counting a pair twice. These are a relaxation: satisfying them does not produce a geometric configuration.

At a search node, already selected parity counts give lower bounds on the z_α , and the numbers of remaining candidates give upper bounds. The available color palette at that node can be smaller than $\mathcal{D}_n$. Enumerating feasible final occupancy vectors once and filtering them as the search proceeds is a useful necessary condition.

For example, an 11-point configuration in G_7 needs 55 colors. The 55th smallest member of $\mathcal{D}_7$ is 68. The stronger modulo-four test excludes every diameter at most 70: the palette through 70 has capacities

$$(b_0, b_1, b_2, b_3) = (14, 17, 17, 8),$$

and no eight nonnegative integers summing to 11 satisfy equations (5) and (6). Thus its largest squared distance must be at least 72. This finite relaxation is reproducible by exhaustive occupancy enumeration and is also checked inside the exact search.

The same finite enumeration gives the following necessary diameter bounds. The middle column counts colors without considering their residues; the last column enforces all four coordinate-parity capacities.

Side length	Target size	Palette threshold	Modulo-four threshold
7	11	68	72
8	13	101	102
9	15	136	149
10	17	181	209

For targets 13, 15, and 17, the independent verifier examines all 77,520, 170,544, and 346,104 nonnegative occupancy vectors, respectively. These thresholds remain necessary conditions only. Applying this inexpensive filter before geometric edge-prefix enumeration avoids spending search time on an arithmetically impossible diameter case.

3.3 A general partition inequality

Proposition 3.2 (Pooled short-distance capacity). *Suppose a selected set is partitioned into q parts, each of squared diameter at most L , and let their sizes be $m_1, \dots, m_q$. Then*

$$\sum_{i=1}^q \binom{m_i}{2} \leq |\mathcal{D}_n \cap [1, L]|, \quad \frac{m^2}{2q} - \frac{m}{2} \leq |\mathcal{D}_n \cap [1, L]|, \quad m = \sum_i m_i.$$

Proof. Every pair within any part consumes a different color in the same short-distance palette. Sum the pairs and apply $\sum_i m_i^2 \geq m^2/q$. $\square$

The palette is shared across *all* parts. Applying an independent copy of the palette budget to each part would discard useful information.

4 An exact algorithm for further terms

4.1 A rainbow clique formulation

Color the edge $\{x, y\}$ of the complete graph on G_n by $\|x - y\|^2$. We seek the largest clique with no repeated edge color. An ordinary clique problem on the points alone does not encode this: the obstruction can involve either three or four vertices.

The search state consists of a selected set A , its used color set U , and candidate points C . For $v \in C$, store the radial colors

$$R_A(v) = \{\|v - a\|^2 : a \in A\}.$$

A candidate is individually admissible exactly when these $|A|$ colors are distinct and avoid U .

Two candidates v, w can belong to a common extension only if

$$R_A(v) \cap R_A(w) = \emptyset, \quad \|v - w\|^2 \notin U \cup R_A(v) \cup R_A(w). \quad (7)$$

Build a compatibility graph using these conditions. Every valid extension is a clique in this graph. The converse need not hold, because different edges between future candidates may still have equal colors. The graph therefore supplies an upper bound, while recursion enforces the full rainbow condition.

4.2 Sound bounds inside recursion

If r additional vertices are needed, the following tests are necessary.

1. There must be at least r candidates.
2. The union of radial colors must have at least $|A|r$ elements. Those selected-to-future edges are all required and all differently colored.
3. The union of used colors, candidate radial colors, and colors of compatible candidate edges must contain at least $\binom{|A|+r}{2}$ colors.
4. A proper coloring of the compatibility graph must use at least r colors. Every clique has at most one vertex in each color class.
5. The parity and coordinate-parity capacities must permit a final occupancy vector of size $|A| + r$.

The later implementation also repeatedly removes vertices whose current compatibility degree is less than $r - 1$. Such a vertex cannot belong to an r -clique. Iterating this deletion is the usual graph-core bound and remains a necessary-condition test.

Greedy coloring is sufficient for correctness. It need not find the chromatic number: any proper coloring proves a valid clique upper bound. The implementation sorts and colors candidates, then branches in reverse color order. Prefix color bounds allow whole remaining prefixes to be discarded.

After selecting v , replace U by $U \cup R_A(v)$. Keep only compatible candidates, add their squared distance to v to their radial sets, and recompute the necessary bounds. All these operations use integer bit masks.

Theorem 4.1 (Correctness of the recursive search). *Provided the initial cases cover every valid configuration up to the declared symmetries, an exhaustive run of this recursion returns a valid set of the target size or proves that no such set exists.*

Proof. The invariant says that A is valid and every retained candidate can individually extend it. Conditions equation (7) are necessary for two candidates to appear together. The color-union and occupancy tests count distinct colors that any extension must use. The clique-coloring test follows from proper coloring. Thus no pruning rule removes a valid completion.

For a branching vertex v , the branch containing v retains all completions that contain it; the subsequent branches exclude v . Induction over the ordered candidate list covers every possible completion once within that search state. At target size the invariant proves validity. If all branches terminate unsuccessfully without a time limit, no completion exists. $\square$

4.3 A unique longest edge gives a small complete case split

Let the target be k , and write the palette in increasing order as $d_1 < d_2 < \dots < d_{d_n}$. A valid k -point configuration has a *unique* longest edge and its squared length is at least

$$d_{\binom{k}{2}}.$$

Enumerate all unordered grid edges meeting this threshold, quotient them by the 48 signed coordinate permutations preserving the cube, and use one representative of each orbit as an initial selected pair.

If its squared length is D , all remaining distances must be *strictly* less than D . This reduces both the candidate list and the color palette. It is stronger than merely anchoring one point. For $n = 7, k = 11$, the threshold is 68; there are 810 eligible grid edges but only 31 orbits under the cube isometry group.

The isometries stabilizing an initial unordered edge can further reduce the first remaining point. Choose its least representative under that stabilizer and require later candidates not to violate this choice. This is applied only with the explicit stabilizer of the current edge; arbitrary sorting of point coordinates independently would be invalid.

The global case-split proof is simple: every target set has exactly one longest edge; a cube isometry sends that edge to one of the enumerated representatives; every other edge remains strictly shorter. A completed case record for every orbit is therefore an impossibility certificate for the target size.

4.4 Anchoring several largest edges

An additional program extends this idea to a prefix of several largest edges; the completed runs use prefixes of up to six. If a target configuration has $K = \binom{k}{2}$ edges and its squared lengths are decreasingly ordered as $D_1 > \dots > D_K$, then

$$D_j \geq d_{K-j+1}.$$

After fixing the first $j - 1$ edges, enumerate the next edge subject to that lower bound and the strict upper bound D_{j-1} . Its endpoints may be among the already selected vertices or among the remaining candidates; both possibilities must be included.

Some pairs among already selected endpoints have not yet entered the ordered prefix. The next length must also be at least the largest such unprocessed length. After choosing D_j , reject the branch if a pair of selected endpoints has length greater than D_j that was not already one of the processed prefix edges. Keep future candidates only when all their distances to selected endpoints are smaller than the active bound and avoid used colors.

The available symmetry group is now the subgroup fixing every processed unordered edge. It preserves the selected endpoint set and the candidate conditions. Taking one representative for the next edge under this subgroup preserves completeness. When the desired prefix length is reached, the ordinary rainbow-clique recursion finishes the search. Induction on the ordered edge prefix proves that every target configuration is covered.

This adds geometric information early, often with fewer arbitrary vertex choices. The effect is strongly instance dependent. The delivered sources include both the original single-diameter version used for the first complete $n = 7$ result and the longer-prefix version, so the original computation remains reproducible.

4.5 Independent solver encodings

An alternative formulation has binary vertex variables x_v and binary edge variables y_{uv} . It suffices to impose

$$y_{uv} \geq x_u + x_v - 1, \quad \sum_{\substack{u < v \\ \|u-v\|^2=d}} y_{uv} \leq 1 \quad (d \in \mathcal{D}_n), \quad \sum_v x_v \geq k.$$

The usual upper inequalities $y_{uv} \leq x_u, x_v$ may also be added. For satisfiability they are optional: unused edge variables can be zero. This is an exact pseudo-Boolean or integer-programming formulation.

A direct SAT encoding uses the implication $(x_u \wedge x_v) \Rightarrow y_{uv}$, an at-most-one constraint on edge variables of each color, and a cardinality lower bound on the vertices. Alternatively one can forbid the union of the endpoints of each pair of equally colored edges. That gives clauses of size three or four but potentially far more clauses.

For future large computations, a proof-producing SAT or pseudo-Boolean solver is attractive: it can emit an independently checkable unsatisfiability trace. The supplied C++ result is an exhaustive-search certificate backed by source, case records, and rerun instructions; it is not presented as a formally checked SAT proof.

4.6 Complexity and practical scope

The palette and distance matrix are inexpensive relative to the exact search. The matrix has n^6 entries; for $n \leq 10$ it is small enough for the delivered implementation. The search remains exponential in the worst case. No polynomial- or quasi-polynomial-time algorithm is claimed.

The C++ program uses fixed masks sized for $n \leq 10$, validates that range, and searches one target size. The formulation and correctness proof extend to larger n ; increasing the mask capacity or using dynamically sized bitsets is an implementation change, not an asymptotic speedup. Palette computations and point-list verification are not subject to this search limit.

5 Computational evidence and reproducibility

5.1 The new exact term at side length seven

An exhaustive run of the unique-longest-edge algorithm rules out 11 points in G_7 . It completed all 31 diameter-edge orbits, visiting 60,290,915 recursive search nodes in 525.875 seconds. The original source snapshot, its SHA-256 digest, the full case log, and the structured result are included. An independent Python enumeration regenerated all 810 eligible grid edges and their 31 orbits and checked that the log covers every case with the correct endpoints and squared diameter.

A matching 10-point witness is

$$A_7 = \{(0, 0, 0), (2, 0, 0), (0, 6, 0), (6, 1, 2), (4, 2, 0), \\ (6, 1, 3), (3, 6, 6), (5, 1, 5), (6, 4, 6), (4, 5, 5)\}.$$

Its 45 pair distances are all distinct. Therefore

$$\boxed{a_7 = 10}, \quad (a_0, \dots, a_7) = (0, 1, 3, 4, 6, 7, 9, 10).$$

The independent case-coverage check verifies the completeness of the outer split and the witness arithmetic. The impossibility conclusion also relies on the correctness of the supplied exhaustive-search program, whose pruning rules are proved in Section 4. It is a reproducible computer-assisted result, not a claim that a short hand enumeration or a formally verified proof trace was obtained.

A second complete computation, using a prefix of the four largest edges together with the later graph pruning, again excluded all 31 cases. It required 5,244,129 calls to the inner clique search and 43.767 seconds. The outer edge-prefix recursion is not included in that inner-call count. The two implementations share core ideas and are not wholly independent proof engines, but they use different complete case refinements and agree on the result. Their recorded times are observations under shared CPU load.

5.2 The new exact term at side length eight

The six-largest-edge prefix algorithm excludes 13 points in G_8 . There are 87 available squared distances. A 13-point set would use 78 of them, so its longest edge must have squared length at least 101. The 660 eligible grid edges form 23 orbits under the 48 cube isometries. Every orbit was exhausted, with 38,430,195 calls to the inner clique search in 445.028 seconds. As above, the node count excludes calls to the outer edge-prefix recursion. Independent enumeration checks all orbit representatives, the logged endpoints and diameters, complete coverage, and the frozen source digest.

The 12-point construction displayed below supplies the matching lower bound. Consequently,

$$\boxed{a_8 = 12}, \quad (a_0, \dots, a_8) = (0, 1, 3, 4, 6, 7, 9, 10, 12).$$

This computation uses the same audited recursive invariants as the side-seven search. The archived proof source accepts prefixes up to six edges; the recursion itself is the same as the version tested against an independent reference enumerator on small instances. The validation package distinguishes those finite regression tests from the mathematical completeness argument and the full side-eight run.

Further budgeted runs attempted to exclude 15 points at side nine and 17 points at side ten. Both reached approximately 600 seconds with status UNKNOWN. The side-nine run completed cases 0 through 8 and stopped during case 9; the side-ten run completed cases 0 through 3 and stopped during case 4. These partial records are retained separately for reproducibility. They give no improved global upper bound, so the final intervals remain $13 \leq a_9 \leq 15$ and $14 \leq a_{10} \leq 17$.

5.3 Explicit improvements at side lengths 8, 9, and 10

The following point lists use coordinates from 0 through $n - 1$. Their complete squared-distance inventories are supplied as JSON.

$$\begin{aligned} A_8 = \{ & (7, 4, 0), (2, 0, 1), (7, 0, 0), (0, 7, 7), \\ & (2, 5, 7), (7, 3, 0), (7, 7, 7), (1, 5, 6), \\ & (4, 7, 3), (3, 1, 7), (5, 0, 0), (4, 1, 1) \}. \end{aligned}$$

The set A_8 has 12 points and 66 distinct squared distances.

$$\begin{aligned} A_9 = \{ & (8, 7, 8), (4, 0, 2), (0, 0, 8), (2, 0, 8), \\ & (8, 8, 0), (8, 8, 8), (4, 8, 0), (0, 2, 6), \\ & (2, 5, 1), (3, 4, 2), (5, 0, 8), (3, 3, 1), (0, 1, 1) \}. \end{aligned}$$

The set A_9 has 13 points and 78 distinct squared distances.

$$\begin{aligned} A_{10} = \{ & (6, 9, 0), (9, 3, 6), (9, 9, 0), (0, 0, 1), \\ & (1, 0, 9), (0, 9, 4), (9, 8, 1), (8, 3, 6), \\ & (2, 1, 8), (7, 7, 0), (2, 5, 9), (6, 0, 9), \\ & (9, 8, 3), (3, 7, 0) \}. \end{aligned}$$

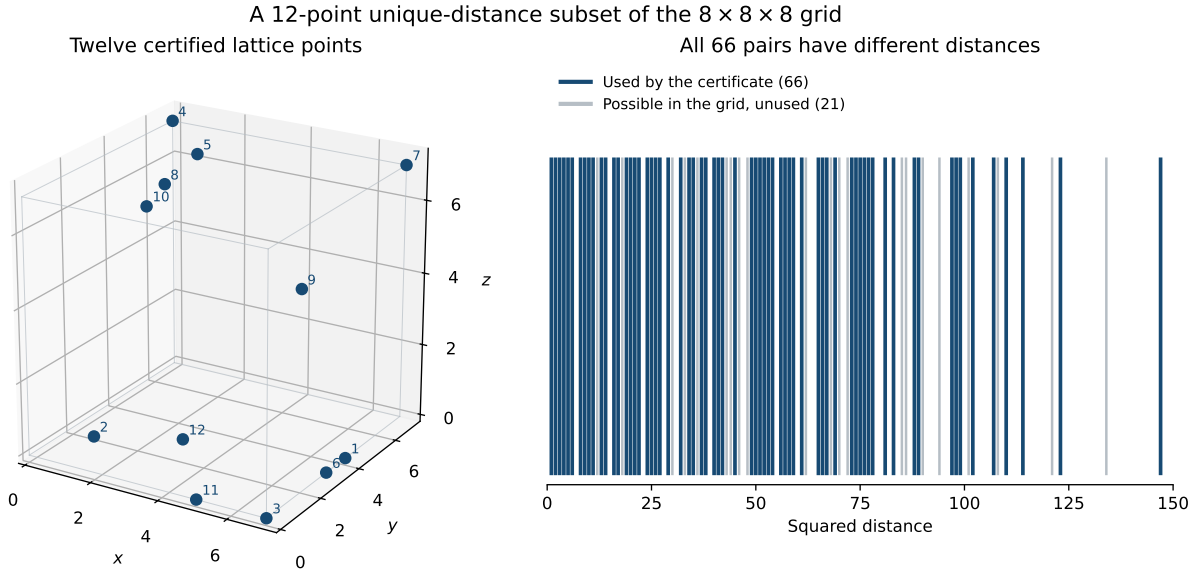


Figure 1: The 12-point witness in G_8 , with its exact squared-distance spectrum. Gray marks are available cube distances that the witness does not use. The 66 blue marks correspond to its 66 unordered pairs.

The set A_{10} has 14 points and 91 distinct squared distances.

5.4 A search that walks through valid configurations

Collision-minimization searches found configurations with only one or two repeated distances, but several stalled there. A more successful procedure maintains a valid k -point set throughout. Delete three points, or four with probability one third, leaving a fixed core. List all grid points that are individually admissible against that core, shuffle their order, and recursively refill while preserving distinctness. A refill with one extra point proves a larger lower bound. An unsuccessful bounded traversal can select a valid refill of the original size as the next state.

This is a heuristic exploration of valid sets; a node cap limits each move. The validity of every completed construction is exact, whereas failure of a bounded traversal has no global meaning. With $C \leq n^3$ remaining candidates and a traversal cap of B successful partial refills, a safe per-move bound is $O(n^3k + B C k)$. Counting only successful partial refills does not count all failed candidate trials, so a claimed $O(Bk)$ bound would be unjustified.

The three displayed witnesses were found from valid starting sets in 497902, 109517, and 5858 moves respectively, with seeds 888112, 999123, and 101314. Recorded wall times were approximately 108.75, 20.41, and 1.29 seconds. These are warm-started search observations, not a controlled comparison of algorithms or machines. The starting sets, source code, and discovery metadata are included in the package. A later version deletes four or five points per move. It produced 16 points in G_{12} and 18 points in G_{14} , completing the certified inequality $a_n \geq n + 4$ for every $8 \leq n \leq 14$. The immutable starting configurations and exact iteration budgets for these runs are included as well.

5.5 Certificate checks

A separate Python verifier checks coordinate ranges, distinct points, and a dictionary from squared distance to its unique unordered point pair. It also verifies the full supplied distance inventory. Such a check takes $O(k^2)$ exact arithmetic operations for a witness of size k . No comparison of rounded Euclidean lengths is used.

5.6 The complete finite catalogue through side length thirty

All lower bounds in Table 2 have explicit coordinate certificates. The upper bounds shown there are the elementary palette and parity bounds; no exactness is asserted for these larger sizes.

Table 2: Certified bounds beyond side length ten. All point lists are supplied.

n	Lower	Upper	n	Lower	Upper
11	15	19	21	22	38
12	16	21	22	23	40
13	17	23	23	24	42
14	18	24	24	25	43
15	18	26	25	26	46
16	18	28	26	26	48
17	19	30	27	27	50
18	20	32	28	28	52
19	21	34	29	29	54
20	22	36	30	30	55

6 Arithmetic density without a three-square sufficiency theorem

For the upper bounds it is enough to use the arithmetic superset

$$\mathcal{B} = \mathbb{N}_{>0} \setminus \{4^a(8b+7) : a, b \in \mathbb{N}\}.$$

Every positive sum of three integer squares belongs to $\mathcal{B}$. To see this, note that a square modulo 8 is 0, 1, or 4, and three such squares cannot sum to 7 modulo 8. If a sum of three squares is divisible by 4, reducing modulo 4 shows that all three coordinates are even. Divide by 4 and repeat. The converse assertion, namely that every member of $\mathcal{B}$ is a sum of three squares, is not needed here.

Lemma 6.1 (Admissible-integer counting). *Let $B(X) = |\mathcal{B} \cap [1, X]|$. Then*

$$B(X) = \frac{5}{6}X + O(\log(2+X)). \quad (8)$$

For every fixed continuously differentiable $f : [0, 3] \rightarrow \mathbb{R}$,

$$\sum_{\substack{r \in \mathcal{B} \\ r \leq 3N^2}} f(r/N^2) = \frac{5}{6}N^2 \int_0^3 f(t) dt + O_f(\log(2+N)). \quad (9)$$

Proof. The excluded progressions $4^a(8b+7)$ are disjoint, since the exact power of 4 dividing such a number identifies a . For $X \geq 1$,

$$B(X) = \lfloor X \rfloor - \sum_{a \geq 0} \left\lfloor \frac{\lfloor X/4^a \rfloor + 1}{8} \right\rfloor.$$

There are $O(\log(2+X))$ nonzero terms. Each is $X/(8 \cdot 4^a) + O(1)$, and the omitted tail of the main geometric series is $O(1)$. Since $\sum_{a \geq 0} (8 \cdot 4^a)^{-1} = 1/6$, this proves equation (8). Partial summation, with the derivative of $f(t/N^2)$ contributing a factor N^{-2} , gives equation (9). $\square$

The elementary arithmetic upper bound is consequently

$$\binom{a_n}{2} \leq B(3(n-1)^2) = \frac{5}{2}(n-1)^2 + O(\log n), \quad a_n \leq \sqrt{5}(n-1) + O(1).$$

This bounds the exact cube palette by an arithmetic superset. It does not assert an asymptotic equality for d_n .

6.1 A limitation of the modulo-four counting relaxation

The leading counts of members of $\mathcal{B} \cap [1, 3N^2]$ in residues $0, 1, 2, 3 \pmod{4}$ are

$$\left(\frac{5}{8}, \frac{3}{4}, \frac{3}{4}, \frac{3}{8} \right) N^2.$$

For residue zero this follows by dividing by four and applying lemma 6.1; residues one and two have density $1/4$ each, and the allowed odd residue-three integers are precisely $3 \pmod{8}$.

These four arithmetic capacities alone cannot improve the leading coefficient $\sqrt{5}$. To see this, put

$$u = \frac{1}{\sqrt{10}}, \quad v = \frac{1}{\sqrt{5}}, \quad p_{000} = \frac{1+4u+3v}{8}, \quad p_{111} = \frac{1-4u+3v}{8},$$

and give each of the other six classes mass $(1-v)/8$. These are positive masses summing to one. For two independent parity classes with this distribution, direct substitution gives Hamming-distance probabilities

$$(P_0, P_1, P_2, P_3) = \left(\frac{1}{4}, \frac{3}{10}, \frac{3}{10}, \frac{3}{20} \right).$$

At $m \sim \sqrt{5}N$, the leading requirements $\frac{1}{2}m^2P_r$ equal the four capacities above. For any fixed $0 < c < \sqrt{5}$, choose integer occupancies of total size $m = \lfloor cN \rfloor$ approximating these proportions. Rounding changes each requirement by $O(N)$, while the capacity slack is $\Theta(N^2)$, so every constraint is eventually satisfied. This constructs feasible arithmetic occupancies. Geometric realizability, and the effect of the exact bounded-coordinate palette, require additional information.

7 Positive-definite kernels turn geometry into a stronger bound

Fix $\lambda > 0$. For probability measures μ, η on the unit cube, define

$$K_\lambda(x, y) = e^{-\lambda \|x-y\|^2}, \quad \mathcal{E}_\lambda(\mu, \eta) = \iint K_\lambda(x, y) d\mu(x) d\eta(y), \quad \mathcal{E}_\lambda(\mu) = \mathcal{E}_\lambda(\mu, \mu).$$

Lemma 7.1 (Gaussian positivity). *Every finite signed measure σ on $[0, 1]^3$ satisfies $\mathcal{E}_\lambda(\sigma) \geq 0$.*

Proof. Use the uniformly convergent expansion on the compact product cube

$$K_\lambda(x, y) = e^{-\lambda\|x\|^2} e^{-\lambda\|y\|^2} \sum_{\beta \in \mathbb{N}^3} \frac{(2\lambda)^{|\beta|}}{\beta!} x^\beta y^\beta.$$

Integrating term by term gives

$$\mathcal{E}_\lambda(\sigma) = \sum_{\beta \in \mathbb{N}^3} \frac{(2\lambda)^{|\beta|}}{\beta!} \left(\int e^{-\lambda\|x\|^2} x^\beta d\sigma(x) \right)^2 \geq 0.$$

□

Theorem 7.2 (Product-potential certificate). *Let ν be a probability measure on $[0, 1]$, with potential and energy*

$$U_\nu(t) = \int e^{-\lambda(t-s)^2} d\nu(s), \quad E_\nu = \int U_\nu(t) d\nu(t).$$

Suppose

$$U_\nu(t) \geq u > 0 \quad (0 \leq t \leq 1), \quad E_\nu \leq e, \quad \kappa = 2u^3 - e^3 > 0.$$

Then every distance-distinct set of m points in G_n , $n \geq 2$, with $N = n - 1$, satisfies

$$\kappa m^2 \leq m + 2 \sum_{r \in \mathcal{D}_n} e^{-\lambda r/N^2}. \quad (10)$$

In particular,

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \left(\frac{5(1 - e^{-3\lambda})}{3\lambda\kappa} \right)^{1/2} \leq \left(\frac{5}{3\lambda\kappa} \right)^{1/2}. \quad (11)$$

Proof. Let $\eta = \nu^{\otimes 3}$. Its potential factors into three one-dimensional potentials, so it is at least u^3 throughout the cube, and $\mathcal{E}_\lambda(\eta) = E_\nu^3 \leq e^3$. Apply lemma 7.1 to $\mu - \eta$:

$$\mathcal{E}_\lambda(\mu) \geq 2\mathcal{E}_\lambda(\mu, \eta) - \mathcal{E}_\lambda(\eta) \geq 2u^3 - e^3 = \kappa.$$

Now take μ to be the empirical probability measure on the selected points after division by N . The diagonal terms each equal 1, and the off-diagonal terms occur twice, giving

$$\kappa m^2 \leq m + 2 \sum_{\{x, y\} \subset A} e^{-\lambda\|x-y\|^2/N^2}.$$

All squared distances are distinct members of $\mathcal{D}_n$, which proves equation (10). Enlarge the palette to $\mathcal{B} \cap [1, 3N^2]$, apply lemma 6.1 with $f(t) = e^{-\lambda t}$, and obtain

$$\kappa m^2 \leq m + \frac{5}{3} N^2 \frac{1 - e^{-3\lambda}}{\lambda} + O(\log N).$$

The already proved estimate $m = O(N)$ allows division by N^2 and passage to the limsup. Finally $N/n \rightarrow 1$. □

The method has two independently checkable inputs. Arithmetic bounds how much total kernel weight distinct squared distances can consume. A probability measure certifies a geometric lower bound on the kernel energy of *every* point distribution in the cube. This is why the argument contains more information than counting available colors.

Proposition 7.3 (Tensorization of the Gaussian energy minimum). *Let $e_1(\lambda)$ be the minimum Gaussian energy over probability measures on $[0, 1]$. The minimum energy over probability measures on $[0, 1]^3$ is exactly $e_1(\lambda)^3$.*

Proof. Continuity of the kernel and compactness give a one-dimensional minimizer ν_* . Varying it toward a point mass at x , the right derivative of the energy at zero is $2(U_{\nu_*}(x) - e_1(\lambda))$, which must be nonnegative. Thus $U_{\nu_*} \geq e_1(\lambda)$ everywhere. The product measure $\nu_*^{\otimes 3}$ has energy $e_1(\lambda)^3$, and the proof of theorem 7.2 with $u = e = e_1(\lambda)$ shows that no cube probability measure has smaller energy. $\square$

This identifies a limitation and an advantage of the method. For a fixed Gaussian kernel, optimizing a one-dimensional measure is enough to obtain the best possible universal energy lower bound. A nonproduct reference measure cannot improve that optimum. Stronger information must come from a different kernel, additional arithmetic or geometric constraints, or a sharper use of the distance palette.

8 A completely analytic certificate

Set

$$\lambda = 4 \log 6, \quad \alpha = \log 6, \quad \nu = \frac{216}{605}(\delta_0 + \delta_1) + \frac{173}{605}\delta_{1/2}, \quad E = \frac{49}{121}.$$

The masses are nonnegative and sum to one.

Lemma 8.1. *For this measure, $U_\nu(x) \geq E$ on $[0, 1]$, with equality at $0, 1/2, 1$. Consequently $E_\nu = E$.*

Proof. Put $t = 2x - 1$. Direct substitution gives

$$605U_\nu(x) = e^{-\alpha t^2} [72 \cosh(2\alpha t) + 173].$$

The required inequality is

$$F(t) := 72 \cosh(2\alpha t) + 173 - 245e^{\alpha t^2} \geq 0 \quad (|t| \leq 1).$$

There is equality at $t = 0$. At $t = 1$,

$$72 \frac{6^2 + 6^{-2}}{2} + 173 = 245 \cdot 6,$$

so $F(1) = 0$ as well.

For $k \geq 1$, the ratio between the positive cosh contribution and the exponential contribution to the coefficient of t^{2k} is

$$r_k = \frac{72}{245} \frac{(2\alpha)^k}{(2k-1)!!}, \quad \frac{r_{k+1}}{r_k} = \frac{2\alpha}{2k+1}.$$

The rational bounds $1.79 < \alpha < 1.80$ imply

$$r_1 = \frac{144\alpha}{245} > 1, \quad r_2 = \frac{96\alpha^2}{245} > 1, \quad r_3 = \frac{192\alpha^3}{1225} < 1.$$

For $k \geq 3$ the ratios continue decreasing. Therefore

$$F(t) = b_1 t^2 + b_2 t^4 - \sum_{k \geq 3} c_k t^{2k}, \quad b_1, b_2, c_k > 0.$$

Absolute convergence and $|t| \leq 1$ give

$$F(t) \geq t^4 \left(b_1 + b_2 - \sum_{k \geq 3} c_k \right) = t^4 F(1) = 0.$$

The support equalities prove $E_\nu = E$. □

For completeness, the two coarse logarithm bounds used in the proof can be checked by rational arithmetic from

$$\log 6 = 2 \sum_{j \geq 0} \frac{(5/7)^{2j+1}}{2j+1}.$$

After the first J terms, the tail is at most

$$\frac{2(5/7)^{2J+1}}{(2J+1)(1 - (5/7)^2)}.$$

Taking $J = 10$ proves $179/100 < \log 6 < 9/5$.

Corollary 8.2. *The analytic bound equation (2) holds.*

Proof. Use $u = e = E$, so $\kappa = E^3$, in theorem 7.2. Since $e^{-3\lambda} = 6^{-12}$, its coefficient is exactly C_6 . □

The proof also identifies an actual Gaussian-energy minimizer on the cube, namely $\nu^{\otimes 3}$, for this value of λ : its energy is E^3 , and no probability measure has smaller energy. This statement concerns a continuous energy problem; it does not construct a large lattice configuration with distinct distances.

9 The stronger four-atom certificate

The stronger certificate uses $\lambda = 10$ and the rational measure

$$\nu = \frac{3183}{10000}(\delta_0 + \delta_1) + \frac{1817}{10000}(\delta_{3791/10000} + \delta_{6209/10000}).$$

All parameters in the verifier are exact integers. Set

$$u = \frac{36532567}{10^8}, \quad e = \frac{36533029}{10^8}. \tag{12}$$

Proposition 9.1 (Verified potential inequalities). *The accompanying integer-arithmetic verification proves*

$$U_\nu(x) \geq u \quad (0 \leq x \leq 1), \quad E_\nu \leq e.$$

Proof. The finite checks and the passage from a mesh to the entire interval are as follows.

Exponential intervals. All mesh and support differences are $d/10000$, $0 \leq d \leq 10000$. To bound $e^{-10(d/10000)^2}$, divide the exponent by 16:

$$y = \frac{d^2}{160000000} \in [0, 5/8].$$

The alternating Taylor sums satisfy

$$S_{21}(y) \leq e^{-y} \leq S_{20}(y), \quad S_j(y) = \sum_{k=0}^j \frac{(-y)^k}{k!}.$$

The terms decrease in magnitude on this interval. Each sum is evaluated outward on a fixed-point scale 10^{40} , and four outward-rounded squarings enclose the original exponential. Every operation is an integer multiplication, division, or comparison.

Potential on a mesh. Evaluate the four-term potential at $x = i/10000$. Reflection symmetry reduces the computation to $0 \leq i \leq 5000$. The smallest certified lower value occurs at $i = 3792$ and is greater than

$$0.36532569867781706035.$$

This displayed decimal is only a readable summary of a rational bound; the verifier does not use decimal rounding to justify an assertion.

Interpolation between mesh points. For $g(x) = e^{-\lambda(x-y)^2}$,

$$|g''(x)| = 2\lambda |2z - 1|e^{-z} \leq 2\lambda, \quad z = \lambda(x - y)^2 \geq 0.$$

Indeed, the maximum of $|2z - 1|e^{-z}$ is 1. Since the weights sum to one, $|U''_\nu| \leq 20$. A twice differentiable function differs from its linear interpolant on an interval of length h by at most $20h^2/8$. Here this is $1/40000000$. Subtracting that quantity from the mesh lower bound gives more than

$$0.36532567367781706035 > u.$$

Thus the check covers every real x , not just the sampled mesh.

Energy. The 16 weighted support-to-support kernel terms, evaluated with upper exponential intervals, give

$$E_\nu < 0.36533028829202422565 < e.$$

The script checks both final rational inequalities exactly. □

Theorem 9.2 (Certified asymptotic coefficient). *Let*

$$\kappa = 2u^3 - e^3 = \frac{48755553510102913673137}{10^{24}} > 0.$$

Then

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n} \leq \sqrt{\frac{1}{6\kappa}} = 1.848895345708975128 \dots < 1.849.$$

Proof. Apply theorem 7.2 with $\lambda = 10$ and proposition 9.1. Dropping the positive factor e^{-30} from $1 - e^{-30}$ only enlarges the upper bound: $5/(3\lambda) = 1/6$. The inequality

$$\frac{1}{6\kappa} < \left(\frac{1849}{1000}\right)^2$$

is an exact rational comparison performed by the verifier. $\square$

The certificate file contains the chosen rational bounds, the exact κ , the interval parameters, and the resulting constant. The verifier is short enough to audit independently and needs only Python's standard library. Floating-point square roots are unnecessary for verification; decimal values are printed only after the rational inequalities have passed.

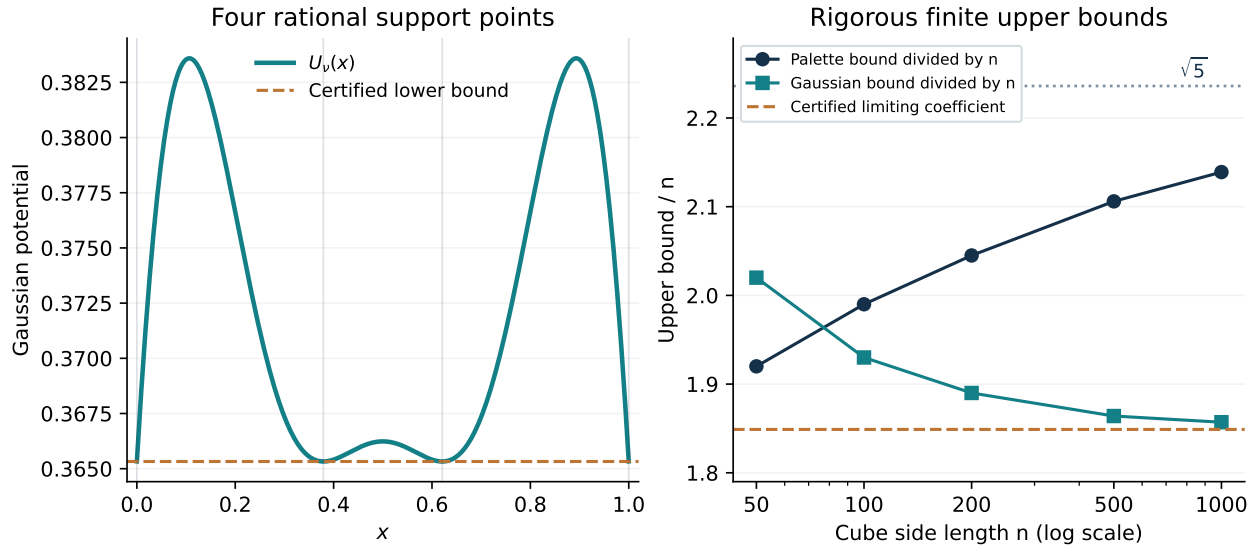


Figure 2: Left: the four-atom potential, illustrated numerically; the proof of its global lower bound is the exact interval certificate. Right: computed finite upper bounds divided by the side length. These curves bound a_n/n ; they are not measured values of that ratio. The dotted line is the elementary coefficient $\sqrt{5}$.

10 Rigorous finite upper bounds from the same certificate

For $N = n - 1$ and $\delta = 10/N^2$, the infinite admissible-distance Laplace sum has the exact identity

$$W_\infty(\delta) = \sum_{r \in \mathcal{B}} e^{-\delta r} = \frac{e^{-\delta}}{1 - e^{-\delta}} - \sum_{a \geq 0} \frac{e^{-7\delta 4^a}}{1 - e^{-8\delta 4^a}}. \quad (13)$$

This follows by subtracting the disjoint forbidden progressions. For any nonnegative integer A , truncating the subtraction after $a = A - 1$ gives an *upper* bound W_A for W_∞ . Enlarging $\mathcal{D}_n$ to all of $\mathcal{B}$ is also an upper bound. Consequently

$$\kappa a_n^2 - a_n \leq 2W_A, \quad a_n \leq \left\lfloor \frac{1 + \sqrt{1 + 8\kappa W_A}}{2\kappa} \right\rfloor. \quad (14)$$

The supplied `finite_bounds.py` evaluates an upper enclosure of W_A entirely with rational arithmetic. It bounds the first geometric series from above and each subtracted series from below. Terms whose initial exponent exceeds 56 are left unsubtracted. Exponentials are reduced by a factor of 128, enclosed with alternating Taylor sums of degrees 31 and 30, and raised by seven directed squarings. The largest integer satisfying the resulting quadratic inequality is located by integer comparisons; numerical root rounding cannot affect the answer.

Table 3: Upper bounds computed with exact arithmetic. The Gaussian column uses equation (14), not an asymptotic expression rounded at finite n .

n	d_n	Palette	Parity	Gaussian
50	4606	96	96	101
100	19798	199	199	193
200	83752	409	409	378
500	554119	1053	1052	932
1000	2287097	2139	2138	1857

The diagonal term a_n in the energy inequality makes this version weak for very small cubes. One should take the minimum of the available bounds. At larger side lengths the Gaussian bound improves the arithmetic palette bound by a substantial margin. For example,

$$a_{100} \leq 193, \quad a_{500} \leq 932, \quad a_{1000} \leq 1857.$$

These are upper bounds, not computed sequence terms. The counting estimate also gives the more precise upper-bound expansion

$$a_n \leq C_{\text{cert}}(n-1) + \frac{1}{2\kappa} + O\left(\frac{\log n}{n}\right).$$

It follows by inserting $2W_\infty(10/(n-1)^2) = (n-1)^2/6 + O(\log n)$ into the quadratic inequality. This is an expansion of an upper bound, not an asymptotic formula for a_n , and its unspecified remainder should not be rounded to obtain finite certified integers.

11 Lower bounds and why the exponent two thirds appears

11.1 An exact finite alteration inequality

For each squared distance d , let H_d be its graph on G_n , with e_d edges and vertex degrees $\deg_d(v)$. Define

$$T = \sum_d \sum_{v \in G_n} \binom{\deg_d(v)}{2}, \quad Q = \sum_d \binom{e_d}{2} - T.$$

Here T counts pairs of equal-length edges sharing a vertex, and Q counts such pairs with disjoint endpoints.

Proposition 11.1 (Sampling and deletion). *For every $0 \leq p \leq 1$,*

$$a_n \geq \left\lceil pn^3 - p^3T - p^4Q \right\rceil.$$

Proof. Retain every grid point independently with probability p . The expected number retained is pn^3 . A conflict with three distinct endpoints survives with probability p^3 , and one with four survives with probability p^4 . Delete one point from each surviving conflict, if that conflict has not already been destroyed. Deletion cannot create a new equality between distances. The final valid set has size at least the number retained minus the number of surviving conflicts. Some outcome is at least its expected lower bound, and the size is integral. $\square$

This is a constructive randomized procedure, although listing all conflicts can be expensive. It is also a useful accounting device: overcounting conflicts weakens the lower bound but does not invalidate it.

11.2 An elementary equal-norm energy estimate

Lemma 11.2. *The number of ordered pairs $u, v \in [-N, N]^3 \cap \mathbb{Z}^3$ with $\|u\|^2 = \|v\|^2$ is $O(N^4)$.*

Proof. Put $a = u - v$, $b = u + v$. This is an injective map into pairs $a, b \in [-2N, 2N]^3$ with $a \cdot b = 0$. Ignoring the parity condition needed to recover u, v only enlarges the count. The case $a = 0$ contributes $O(N^3)$.

For $a \neq 0$, write $a = ga_0$, where a_0 is primitive, $r = \|a_0\|_\infty$, and $1 \leq g \leq 2N/r$. Permute coordinates so that $|(a_0)_3| = r$, and put $h = \gcd((a_0)_2, r)$. Primitivity implies $\gcd((a_0)_1, h) = 1$. The congruence

$$(a_0)_1 b_1 + (a_0)_2 b_2 \equiv 0 \pmod{r}$$

forces $h \mid b_1$. There are at most $4N/h + 1$ possibilities for b_1 . For each, b_2 occupies one residue class modulo r/h , giving at most $4Nh/r + 1$ possibilities. The equation then determines b_3 uniquely, if it is in range. Thus the number of b 's is at most

$$\left(\frac{4N}{h} + 1\right) \left(\frac{4Nh}{r} + 1\right) \leq \frac{16N^2}{r} + 8N + 1.$$

There are at most $(2r+1)^3 - (2r-1)^3 = 24r^2 + 2$ choices for a primitive a_0 of sup norm r . Summing the upper bound

$$\sum_{r=1}^{2N} (24r^2 + 2) \frac{2N}{r} \left(\frac{16N^2}{r} + 8N + 1\right)$$

gives $O(N^4)$. The terms involving $1/r$ or $1/r^2$ are smaller than that order, so no extra logarithm is required. $\square$

Corollary 11.3 (Elementary polynomial lower bound). *There is an absolute $c_0 > 0$ such that $a_n \geq c_0 n^{2/3}$ for all sufficiently large n .*

Proof. Let $r_3(d)$ count displacement vectors of squared norm d in $[-(n-1), n-1]^3$. The preceding lemma gives $\sum_d r_3(d)^2 = O(n^4)$. Since $\deg_d(v) \leq r_3(d)$ and $e_d \leq n^3 r_3(d)/2$,

$$T = O(n^7), \quad Q = O(n^{10}).$$

Choose $p = \gamma n^{-7/3}$, where $\gamma > 0$ is a sufficiently small constant. The alteration bound becomes

$$pn^3 - p^3 T - p^4 Q \geq (\gamma - C\gamma^4)n^{2/3} - C'\gamma^3.$$

For a suitable fixed γ , the leading coefficient is positive. $\square$

The exponent arises from balancing retained vertices against four-vertex conflicts: pn^3 and p^4n^{10} have the same order when $p \asymp n^{-7/3}$. The three-vertex conflicts are of lower order at that scale.

The logarithmic improvement in equation (1) comes from the stronger hypergraph analysis of Lefmann and Thiele [3, 4]. The elementary proof above explains the scale, but it is not an improvement over their theorem. In particular, the newly generated finite configurations do not establish a linear lower bound.

12 Structural properties and monotonicity

12.1 Properties that follow immediately

The sequence is nondecreasing, because $G_n \subseteq G_{n+1}$. It is unbounded by equation (1), or by embedding successively longer finite Golomb rulers on a coordinate line. Consequently there are infinitely many strict increases.

Translations, cube isometries, and nonzero uniform integer dilations preserve distinctness. Restricting a valid configuration to a subbox also preserves it. More generally, for the rectangular-box extremal function $F(l_1, l_2, l_3)$, cutting one coordinate interval gives

$$F(l_1 + l'_1, l_2, l_3) \leq F(l_1, l_2, l_3) + F(l'_1, l_2, l_3).$$

No requirement is imposed on distances crossing the cut when deriving this upper bound, so the inequality is valid.

All nonzero ordered difference vectors of a valid configuration are distinct. In the usual additive terminology it is therefore a Sidon configuration. The converse fails: different displacement vectors, such as $(1, 0, 0)$ and $(0, 1, 0)$, can have the same norm. Likewise, a Cartesian product usually fails because translated copies repeat every internal distance. These observations explain why standard additive constructions require additional work here.

12.2 Limiting measures of linearly sized configurations

Compactness supplies weakly convergent subsequences of the scaled empirical measures. The next result therefore applies to every positive subsequential ratio of configuration size to side length.

Proposition 12.1 (A bounded squared-distance density). *Suppose $n_j \rightarrow \infty$, $N_j = n_j - 1$, and valid configurations $A_j \subseteq G_{n_j}$ have sizes m_j with $m_j/N_j \rightarrow c > 0$. Let their empirical probability measures after scaling by N_j^{-1} converge weakly to μ on the unit cube. If ρ_μ is the law of $\|X - Y\|^2$ for independent X, Y with law μ , then*

$$\rho_\mu \leq \frac{5}{3c^2} dt \quad \text{on } [0, 3].$$

In particular, this squared-distance law has a bounded density and μ has no atoms.

Proof. Write μ_j and ρ_j for the finite empirical measures and their squared-distance laws. Weak convergence on the compact cube passes to product measures and then through the continuous squared-distance map. For nonnegative $f \in C^1[0, 3]$, uniqueness of the used distances gives

$$\int f d\rho_j = \frac{f(0)}{m_j} + \frac{2}{m_j^2} \sum_{d \text{ used}} f(d/N_j^2) \leq \frac{f(0)}{m_j} + \frac{2}{m_j^2} \sum_{\substack{r \in B \\ r \leq 3N_j^2}} f(r/N_j^2).$$

Apply lemma 6.1 and take the limit. The resulting inequality $\int f d\rho_\mu \leq (5/(3c^2)) \int_0^3 f(t) dt$ extends by approximation to every nonnegative continuous function, which proves domination of measures. An atom of μ would give an atom of ρ_μ at zero, so none is possible. $\square$

For example, the proposition implies the uniform local bound

$$\mu(B(x, r)) \leq \sqrt{\frac{20}{3c^2}} r,$$

because two points in that ball are at squared distance at most $4r^2$. A stronger dimensional statement follows from

$$I_s(\mu) := \iint \|x - y\|^{-s} d\mu(x) d\mu(y) \leq \frac{5}{3c^2} \frac{3^{1-s/2}}{1 - s/2} < \infty \quad (0 < s < 2).$$

Every Borel set of positive μ -measure therefore has Hausdorff dimension at least two. Here is the relevant energy argument. The potential $\int \|x - y\|^{-s} d\mu(y)$ is finite at μ -almost every x . Inside any set of positive measure, restrict to a positive-measure subset where that potential is at most some finite M . A ball of radius r centered in this subset has μ -mass at most Mr^s . A covering by balls, enlarged by a factor two to center them in the subset, then has $\sum r_i^s$ bounded away from zero. Its Hausdorff dimension is at least s ; let s increase to two.

There is a useful continuous optimization problem behind this constraint. Define

$$F(\mu) = \inf\{M \geq 0 : \rho_\mu \leq M dt\}, \quad \beta = \inf_{\mu \in \mathcal{P}([0,1]^3)} F(\mu),$$

allowing $F(\mu) = \infty$. Since every squared-distance law has mass one on an interval of length three, $\beta \geq 1/3$. Every positive limiting ratio satisfies

$$c \leq \sqrt{\frac{5}{3\beta}}.$$

For any nonzero nonnegative continuous test function f ,

$$\beta \geq \frac{\inf_{\mu} \iint f(\|x - y\|^2) d\mu(x) d\mu(y)}{\int_0^3 f(t) dt}.$$

The Gaussian argument is one instance of this inequality. Optimizing over several tests or imposing the full density constraint offers a systematic further relaxation. No interchange of infimum and supremum, or strong-duality assertion, is needed here. Discrete configurations approaching a feasible limiting measure would still require a separate construction.

12.3 Strict monotonicity would already be a strong theorem

Proposition 12.2. *If $a_{n+1} > a_n$ for every sufficiently large n , then*

$$a_n \geq n - O(1).$$

Thus eventual strict monotonicity would imply a linear lower bound. If instead $a_n = o(n)$, then among the first N transitions, all but $o(N)$ are plateaus.

Proof. The sequence is integral. Each strict increase is at least one, so eventual strict increase telescopes to $a_n \geq n - O(1)$. Conversely, the number of strict increases up to N is at most $a_N - a_0$; under a sublinear upper bound this is $o(N)$. $\square$

The known asymptotic bounds do not decide between these alternatives. The small computed terms increase, but that observation does not establish strict increase indefinitely. A proof of strict monotonicity would do much more than settle a local-looking sequence question: it would close the known exponent gap in favor of order n .

12.4 What the evidence does and does not suggest

The now-known exact values satisfy

$$a_n = \lfloor 3n/2 \rfloor \quad (0 \leq n \leq 8).$$

This is a finite observation and a reasonable target for the next searches. No argument here supports extrapolating the formula to all n . Such a formula would, in particular, supply a linear lower bound and resolve the exponent question, far beyond the currently established construction theorem.

The small values and the available constructions are useful search targets. They are not enough to select a growth exponent by regression. The interval between the known lower and upper bounds is still wide; finite-size effects and heuristic search quality both affect the observed ratios.

The proper current conclusion from this report is

$$cn^{2/3}(\log n)^{1/3} \leq a_n \leq (1.848895345709 \dots + o(1))n.$$

There is no general formula proved here, no proof of $\Theta(n)$, and no proof of $o(n)$.

13 Further research questions and a concrete next program

1. **Continue exact terms with independently checked impossibility proofs.** The next unproved target sizes in the finite table should be searched using the unique-diameter split. Preserve a record for every case and avoid collapsing partial UNSAT results into a global conclusion. Translate promising instances into SAT or pseudo-Boolean form and request a proof trace that a separate checker can verify.
2. **Optimize complete geometric case splits.** Diameter anchoring is effective because the high-distance palette forces endpoints toward the boundary. The computations already demonstrate the benefit of anchoring several largest edges. The next questions are how to select the prefix length adaptively and how to use its incidence pattern before enumerating endpoints. The covering argument must explicitly allow these edges to share endpoints.
3. **Use arithmetic occupancy beyond modulus four.** Modulo-four coordinate parity has only eight classes and is cheap. Moduli eight, three, or five retain more information but create larger occupancy spaces. A useful goal is a hierarchy that tightens bounds without spending more time on the relaxation than it saves in search. An infeasible modular relaxation should produce a compact arithmetic certificate.
4. **Improve lower-bound search and publish reusable witnesses.** Tabu search, deletion and exact refill, and adaptive vertex moves can be combined with an archive of high-quality configurations. Preserve the full point lists, deterministic iteration budgets, and checksums. Transfer a solution from G_n into G_{n+1} , then search for an extension or a small replacement. Failure of that local neighborhood must remain a local statement.

5. **Optimize the kernel certificate.** Vary λ , the support positions, and the masses in one dimension; then verify every resulting improvement by interval arithmetic or analytic inequalities. Mixtures of Gaussians are positive definite and may better exploit the shape of the admissible distance set. By proposition 7.3, nonproduct reference measures cannot improve the optimum for a single Gaussian. Mixtures no longer have the same product factorization and lead to a different continuous energy problem, where correlations may matter.
6. **Combine kernels with congruence information.** The Gaussian bound uses only the density $5/6$ of an arithmetic superset of distances. Periodic weights or residue-indexed positive semidefinite kernels could incorporate parity capacities directly into the continuous energy problem. Positivity must be proved for the combined kernel, not inferred from a numerical matrix sample. The full bounded-density relaxation in proposition 12.1 provides another systematic target. Determine or bound β , and study whether its minimizing measures can arise as limits of discrete configurations.
7. **Resolve the growth exponent.** Does $a_n = n^{2/3+o(1)}$, does $a_n = n^{1-o(1)}$, or is the exponent intermediate? An upper bound $O(n^{1-\varepsilon})$, or a lower bound $n^{2/3+\varepsilon}$, would represent a substantially larger advance than improving the linear coefficient.
8. **Understand the exact box palette asymptotically.** Determine the deficit between $|\mathcal{D}_n|$ and $B(3(n-1)^2)$. The coordinate cap excludes representations near the boundary, and unrestricted three-square representability alone does not quantify that deficit. A sharp error term would improve finite bounds and clarify how much of the obstruction is arithmetic versus geometric.
9. **Construct the known lower-bound scale explicitly.** The hypergraph existence theorem has a strong asymptotic guarantee, but a usable implementation with explicit constants would make it possible to compare that guarantee with optimization heuristics. Derandomization by conditional expectation is another concrete direction.
10. **Settle strict monotonicity or find the first plateau.** This question should be pursued together with the asymptotic problem, in view of proposition 12.2. A plateau needs an exact upper certificate on the larger cube and a matching witness on the smaller one. A general extension theorem would need to control all new distances simultaneously.

A How to reproduce and inspect the package

The root of the ZIP contains this document's `a275672.tex` and compiled PDF. The `src` directory contains independent certificate verifiers and the exact finite-bound calculator. The `data` directory contains coordinates and verified numerical results. Search source snapshots and retained run records are in `computations/exact`; constructive programs and immutable starting configurations are in `computations/constructions`. The `validation` directory contains the separate source audit and small-instance comparisons against an exhaustive reference enumerator. Its `fresh_audit` subdirectory contains a second independent source review, differential tests, and sanitizer-run records. The `computations/exploratory` directory retains explicitly inconclusive larger-case attempts and the later filtered sources.

The numerical bounds can be regenerated with Python 3:

```
python3 src/verify_results.py
python3 src/verify_gaussian_certificate.py
python3 src/finite_bounds.py 50 100 200 500 1000
```

The first command checks the complete construction catalogue, the diameter-case coverage of the exact runs, arithmetic lemmas, and an independent rational implementation of the potential certificate. The second runs the original rational potential verifier. The third uses rational exponential intervals and prints rigorous finite upper bounds. None requires a numerical optimization library.

The C++ search programs require a C++17 compiler. Full target searches may run substantially longer than witness checks. The README gives the exact invocation and source version for each completed case run. A reported time limit, an interruption, or a partial-case run must never be interpreted as a proof for the entire cube. The validated front end `src/run_exact.py` checks arguments and builds the selected source automatically; for example,

```
python3 src/run_exact.py 8 13 --mode top6 --seconds 3600
```

It also resolves trivially oversized targets before they reach the unchanged historical C++ sources, whose signed-integer pair-count expression assumes an ordinary target within the supported grid.

The PDF can be rebuilt with `latexmk -pdf a275672.tex`. Standard LaTeX packages and the included figure assets suffice; no Internet access is needed for the rebuild. The supplied checksum manifest records the packaged artifacts used for the reported computations.

B An independent geometric bound

This optional argument gives the weaker coefficient

$$C_{\text{geom}} = \frac{15\sqrt{3} - \sqrt{15}}{11} = 2.0097980697\dots$$

It is included because it uses geometric concentration rather than positive-definite kernels.

If $r_1 < \dots < r_s$ are distinct members of $\mathcal{B}$, lemma 6.1 implies

$$\sum_{j=1}^s r_j \geq \frac{3}{5}s^2 - O(s \log(s+2)).$$

Choose a subsequence with $m/N \rightarrow c > 0$; the case $c = 0$ is immediate. For the corresponding empirical measures, normalized to the unit cube, the sum-of-squared-distances identity gives

$$\frac{3}{20}c^2 + o(1) \leq \sum_{i=1}^3 \text{Var}(X_i).$$

Put $w(x) = \sum_i x_i(1 - x_i)$. Completing the square in the coordinate means yields

$$\sum_i \text{Var}(X_i) \leq \frac{3}{4} - \mathbb{E}w(X).$$

For $t < 1/4$, set $r = (1 - \sqrt{1 - 4t})/2$. In a fixed corner octant write $y_i = \min(x_i, 1 - x_i)$ and $s = \sum_i y_i$. Since $y_i \leq 1/2$, $w \geq s/2$; hence $w \leq t < 1/4$ implies $s < 1/2$. Also $w = s - \sum_i y_i^2 \geq s - s^2$, so $s \leq r$. The low- w points in that octant lie in a positive ℓ_1 simplex of radius r , whose squared diameter is at most $2r^2$.

Let K be the number of low- w points across all eight octants. By proposition 3.2 and lemma 6.1,

$$\frac{K^2}{16} - \frac{K}{2} \leq \frac{5}{3}r^2N^2 + O(\log N).$$

Thus $K/N \leq Ar + o(1)$, where $A = \sqrt{80/3}$, uniformly on fixed r -intervals. The error can, for instance, be bounded by $O(\sqrt{N} + \log N/N)$, which suffices for integration.

Write $R = c/A$. The initial estimate $c \leq \sqrt{5}$ gives $R < 1/2$. Integrating the resulting lower bound for the tail of w gives

$$\mathbb{E}w \geq \int_0^R (1 - r/R)(1 - 2r) dr + o(1) = \frac{R}{2} - \frac{R^2}{3} + o(1).$$

Combining the last inequalities and taking the limit yields

$$\frac{11}{80}c^2 + \frac{c}{2A} \leq \frac{3}{4}, \quad \text{or} \quad 11c^2 + 2\sqrt{15}c \leq 60.$$

The positive root is C_{geom} , as asserted.

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